INTELIGÊNCIA ARTIFICIAL

conteúdo extra

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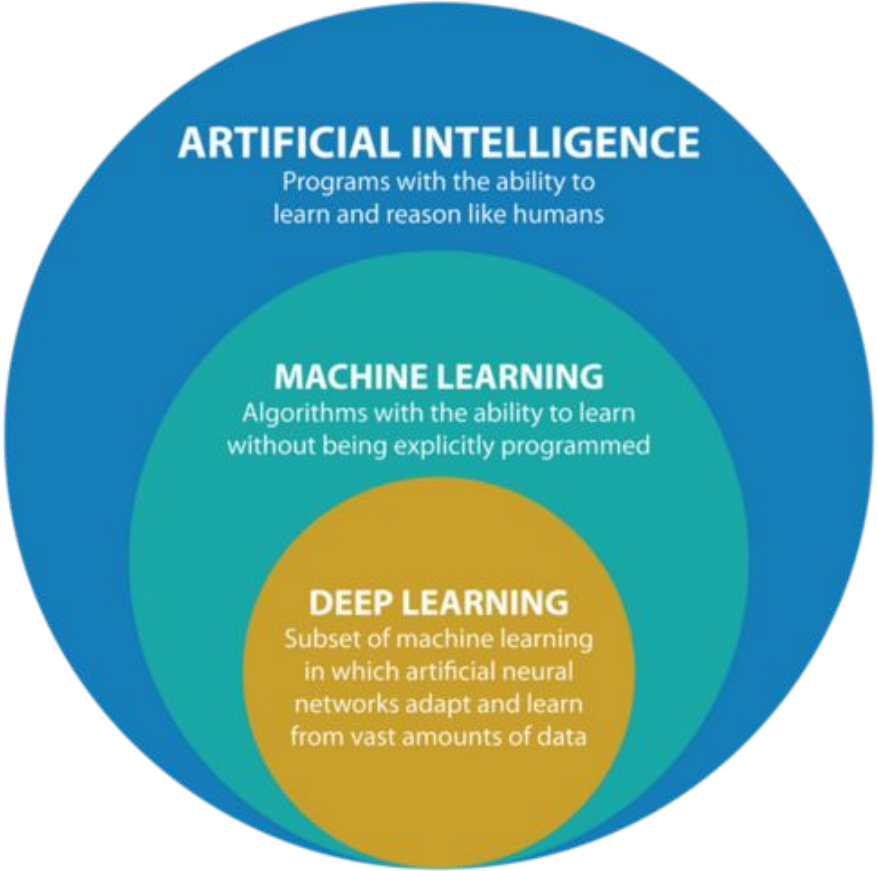
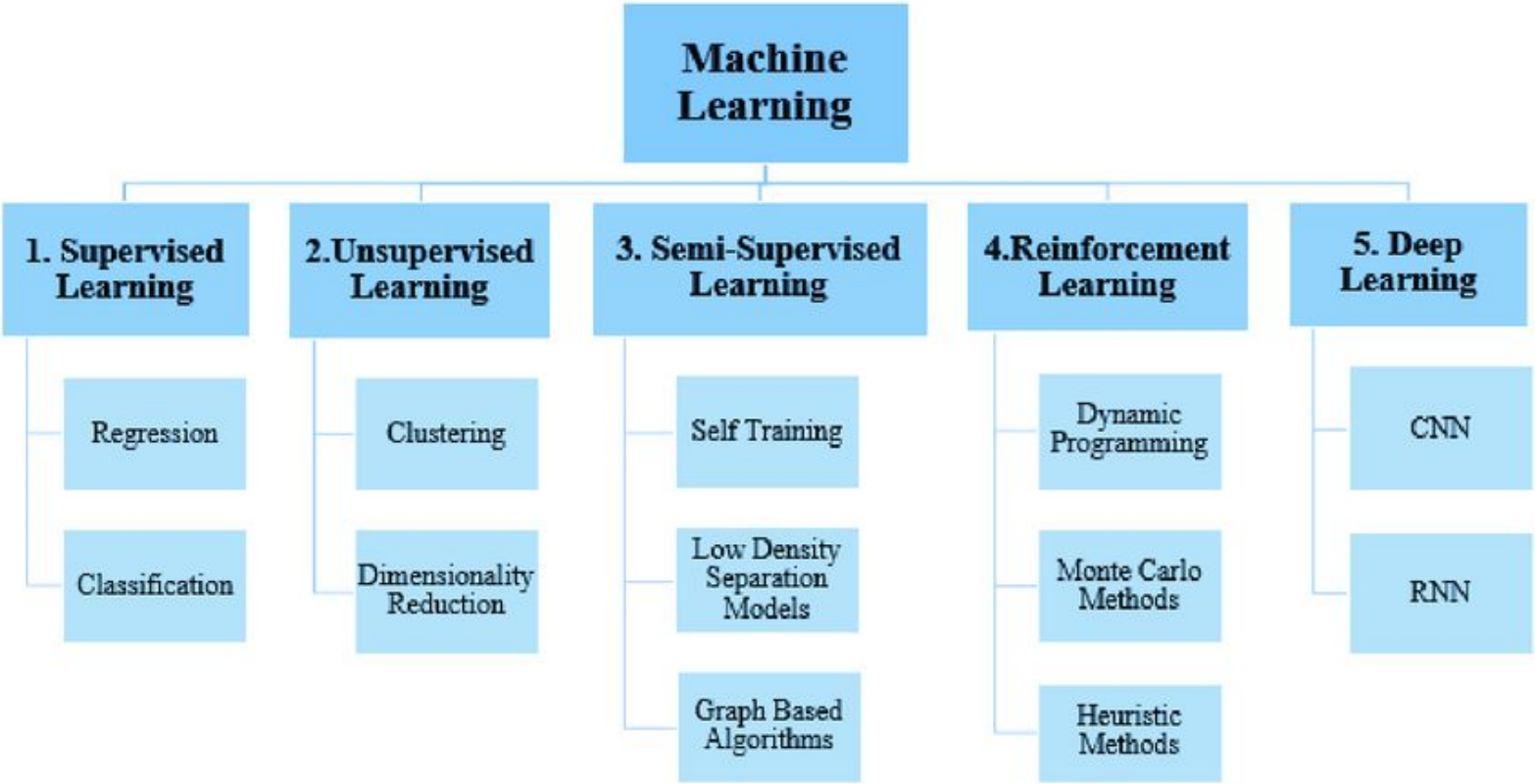
Machine Learning vs IA

Machine Learning vs IA



Machine Learning Areas

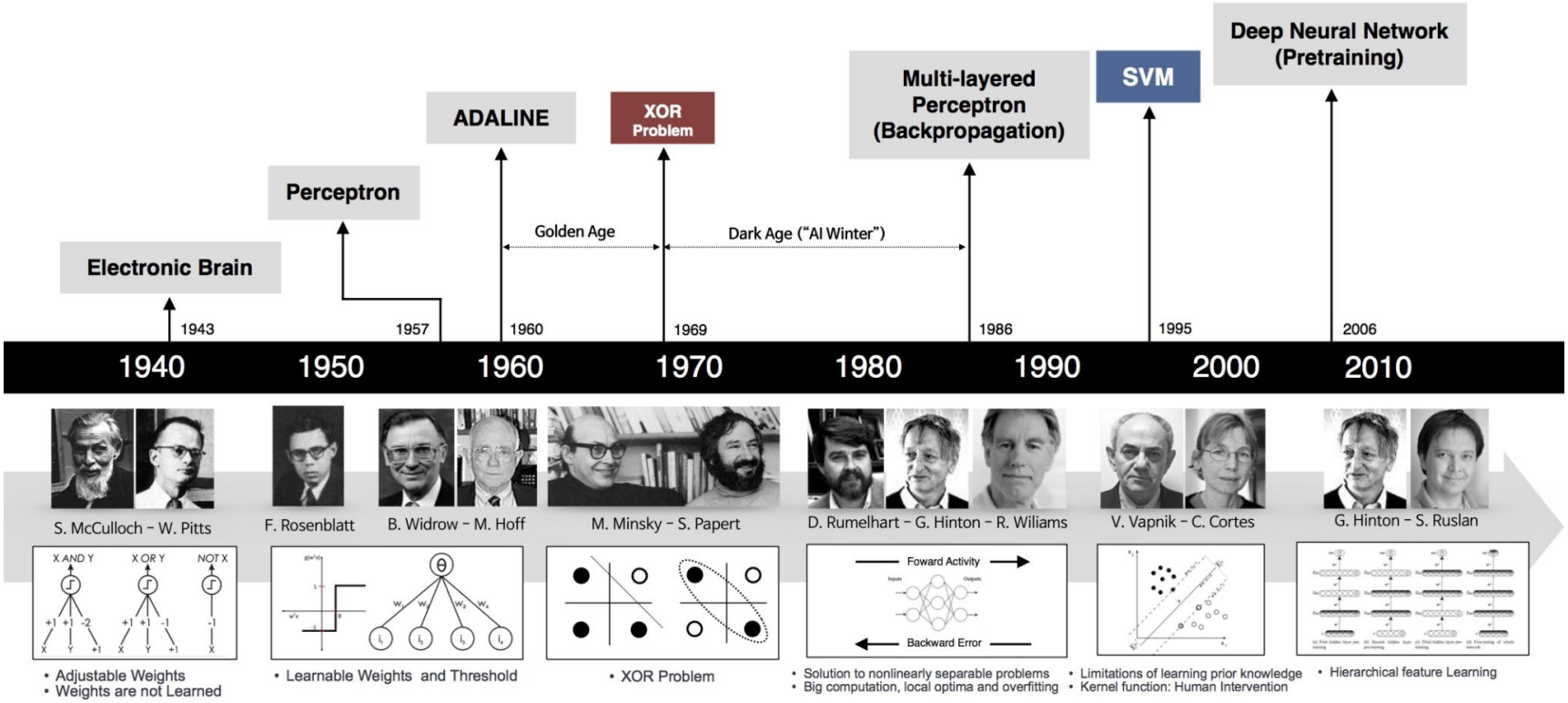


1

Histórico

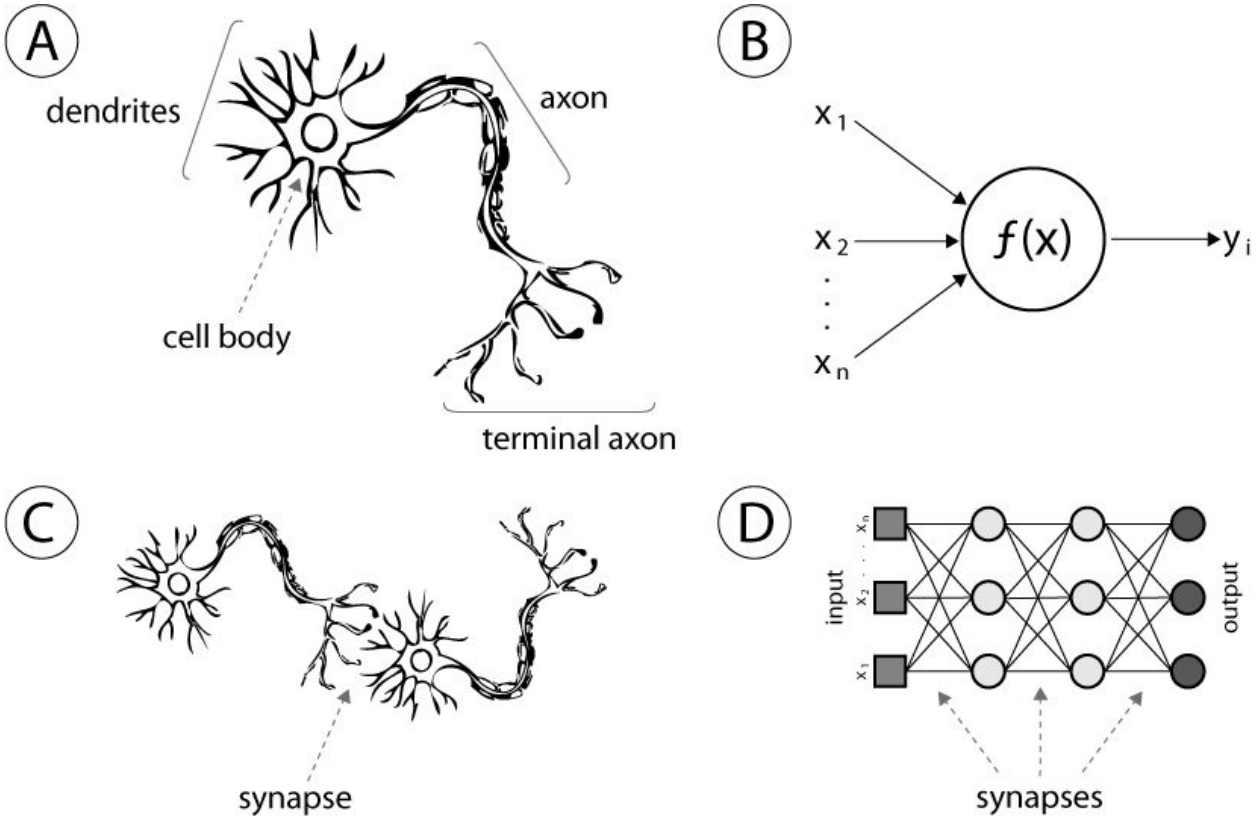
Deep Learning Timeline

Ref: https://www.deeplearningbook.com.br/uma-breve-historia-das-redes-neurais-artificiais/



Redes Neurais Artificiais

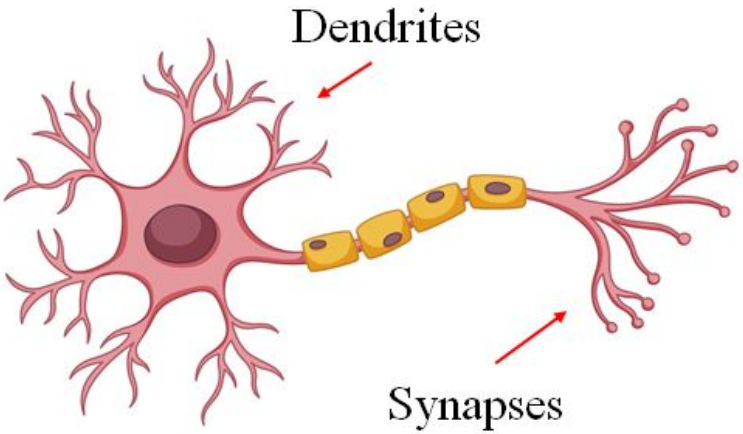
ref: https://medium.com/@ivanliljeqvist/the-essence-of-artificial-neural-networks-5de300c995d6



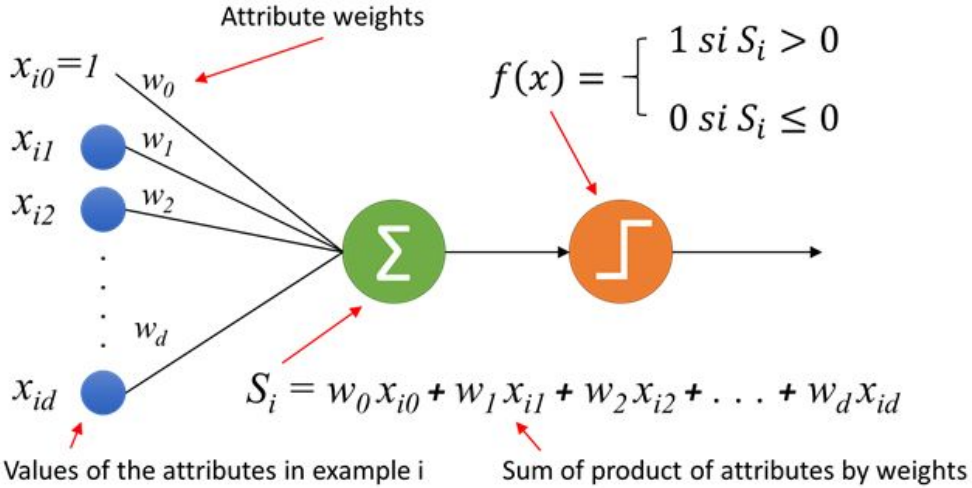
2

Redes Neurais Artificiais

Redes Neurais Artificiais - Perceptron

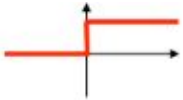
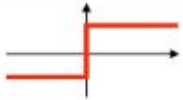
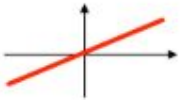
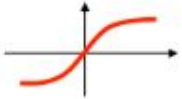


NEURON

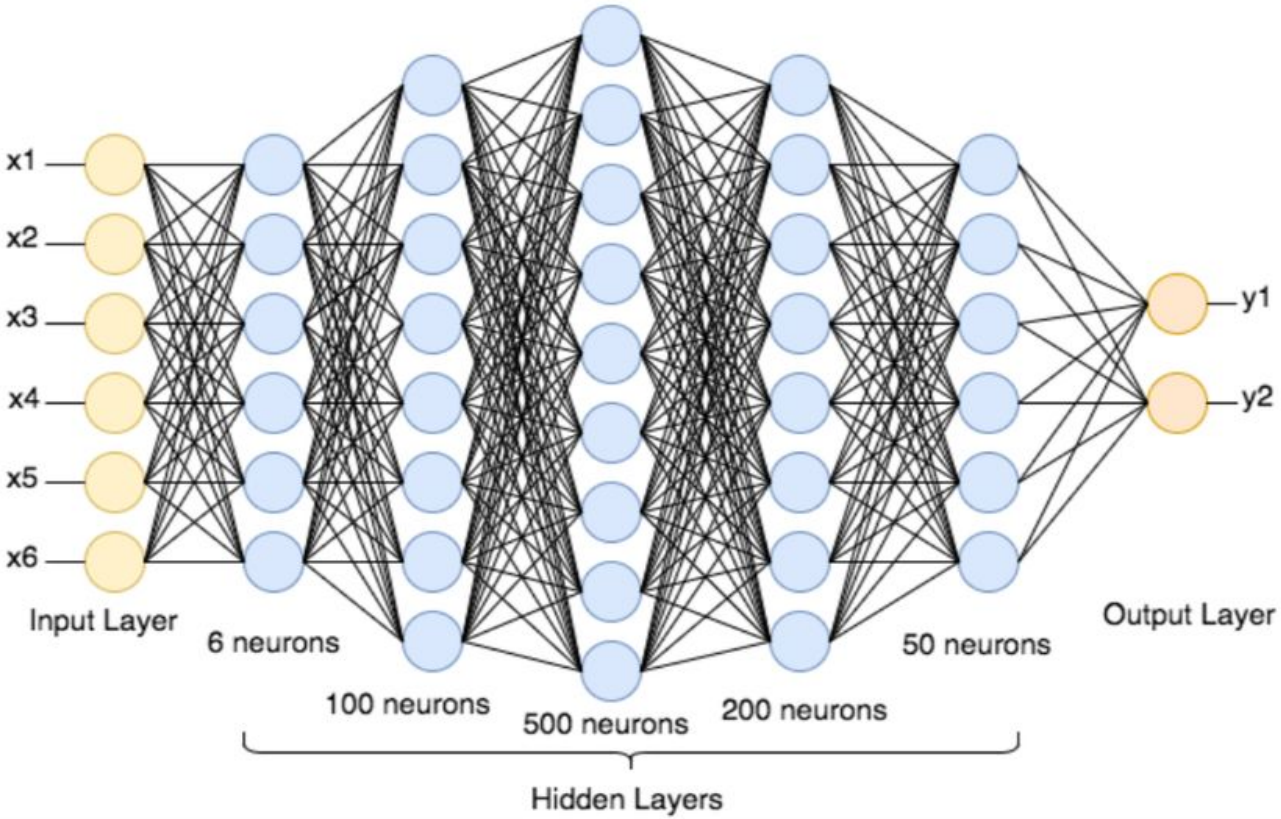


PERCEPTRON

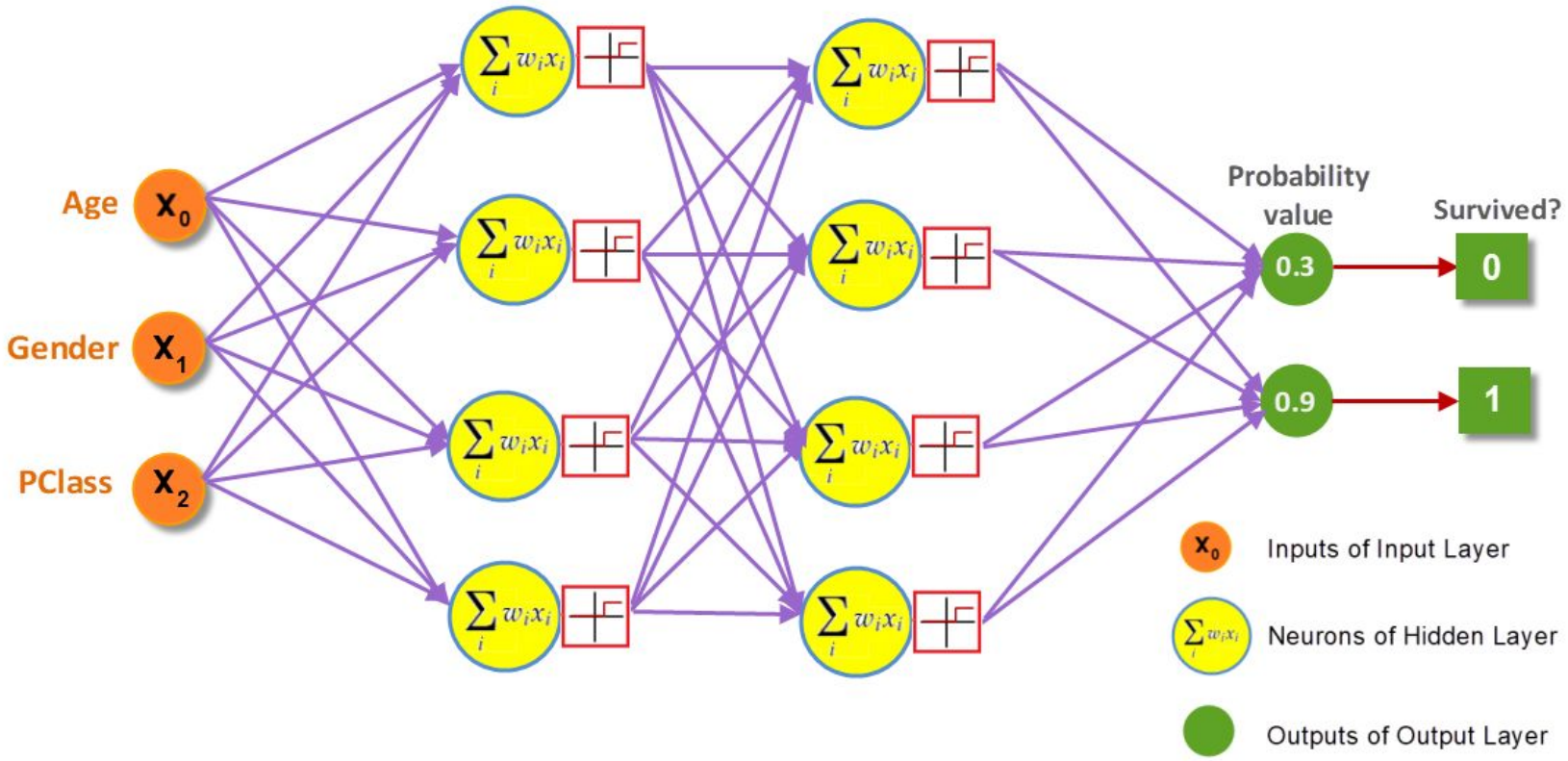
Redes Neurais Artificiais - Funções de Ativação

Unit step (Heaviside)	$\phi(z) = \begin{cases} 0, & z < 0, \\ 0.5, & z = 0, \\ 1, & z > 0, \end{cases}$	Perceptron variant	
Sign (Signum)	$\phi(z) = \begin{cases} -1, & z < 0, \\ 0, & z = 0, \\ 1, & z > 0, \end{cases}$	Perceptron variant	
Linear	$\phi(z) = z$	Adaline, linear regression	
Piece-wise linear	$\phi(z) = \begin{cases} 1, & z \geq \frac{1}{2}, \\ z + \frac{1}{2}, & -\frac{1}{2} < z < \frac{1}{2}, \\ 0, & z \leq -\frac{1}{2}, \end{cases}$	Support vector machine	
Logistic (sigmoid)	$\phi(z) = \frac{1}{1 + e^{-z}}$	Logistic regression, Multi-layer NN	
Hyperbolic tangent	$\phi(z) = \frac{e^{z} - e^{-z}}{e^{z} + e^{-z}}$	Multi-layer Neural Networks	
Rectifier, ReLU (Rectified Linear Unit)	$\phi(z) = max(0, z)$	Multi-layer Neural Networks	

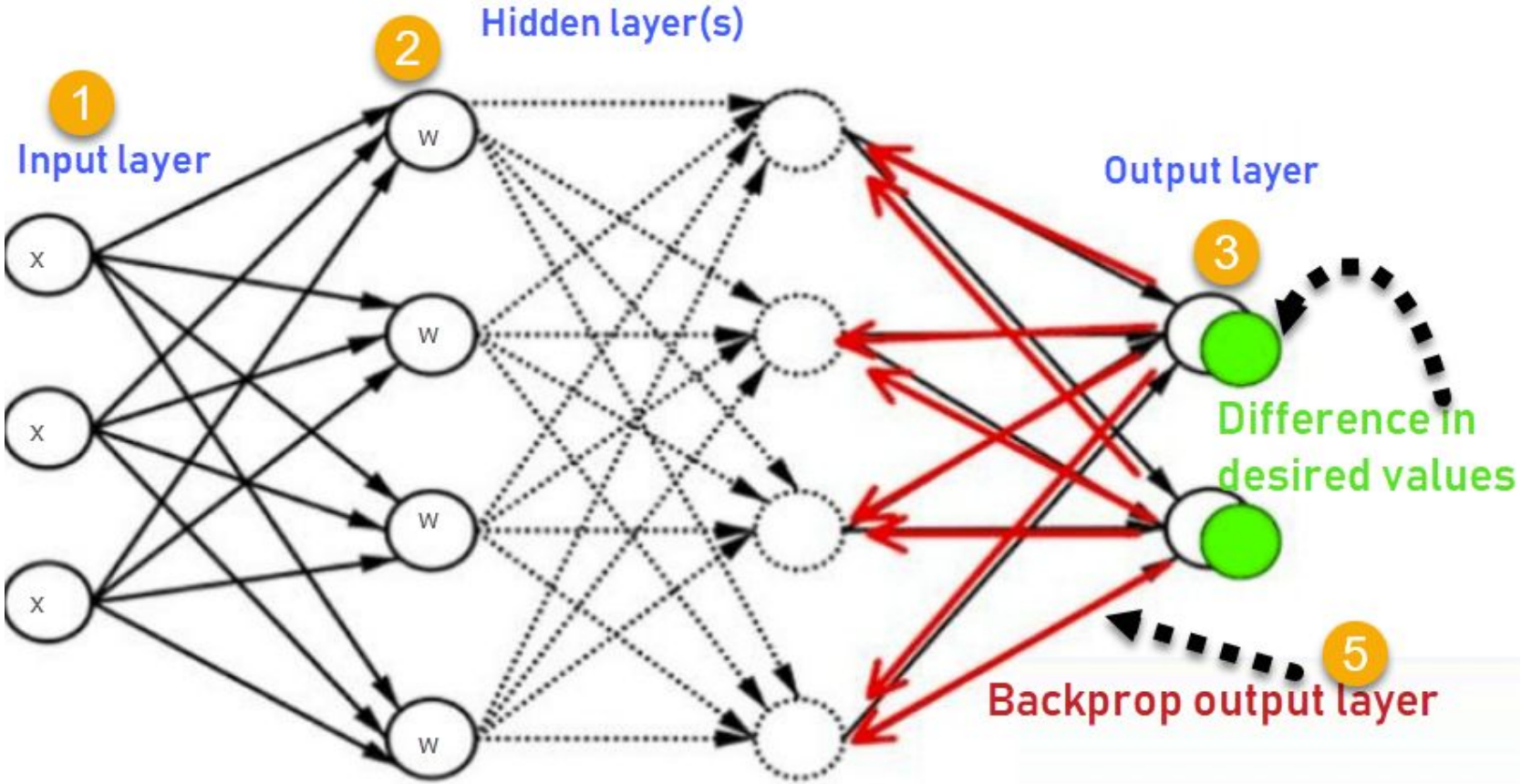
Redes Neurais Artificiais - Multi-Layer



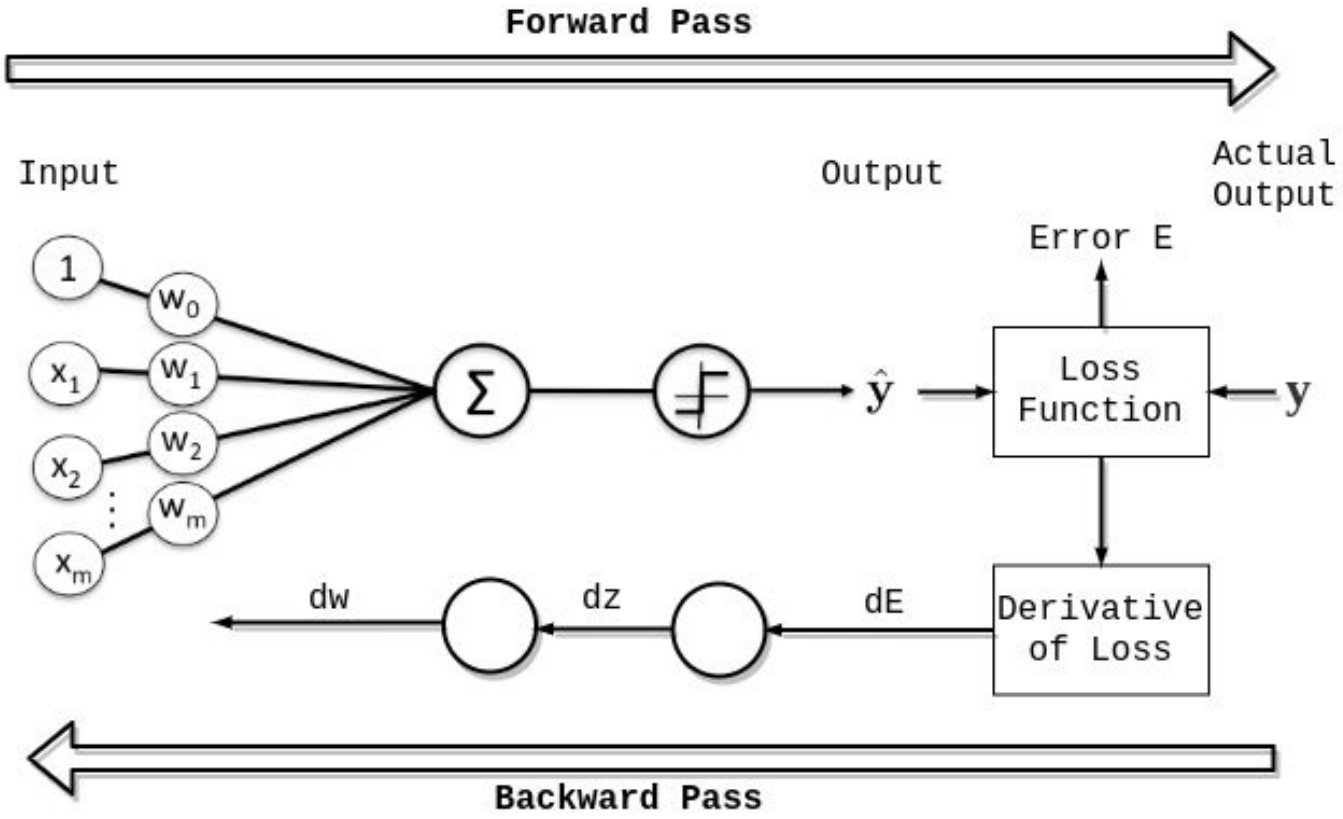
Redes Neurais Artificiais



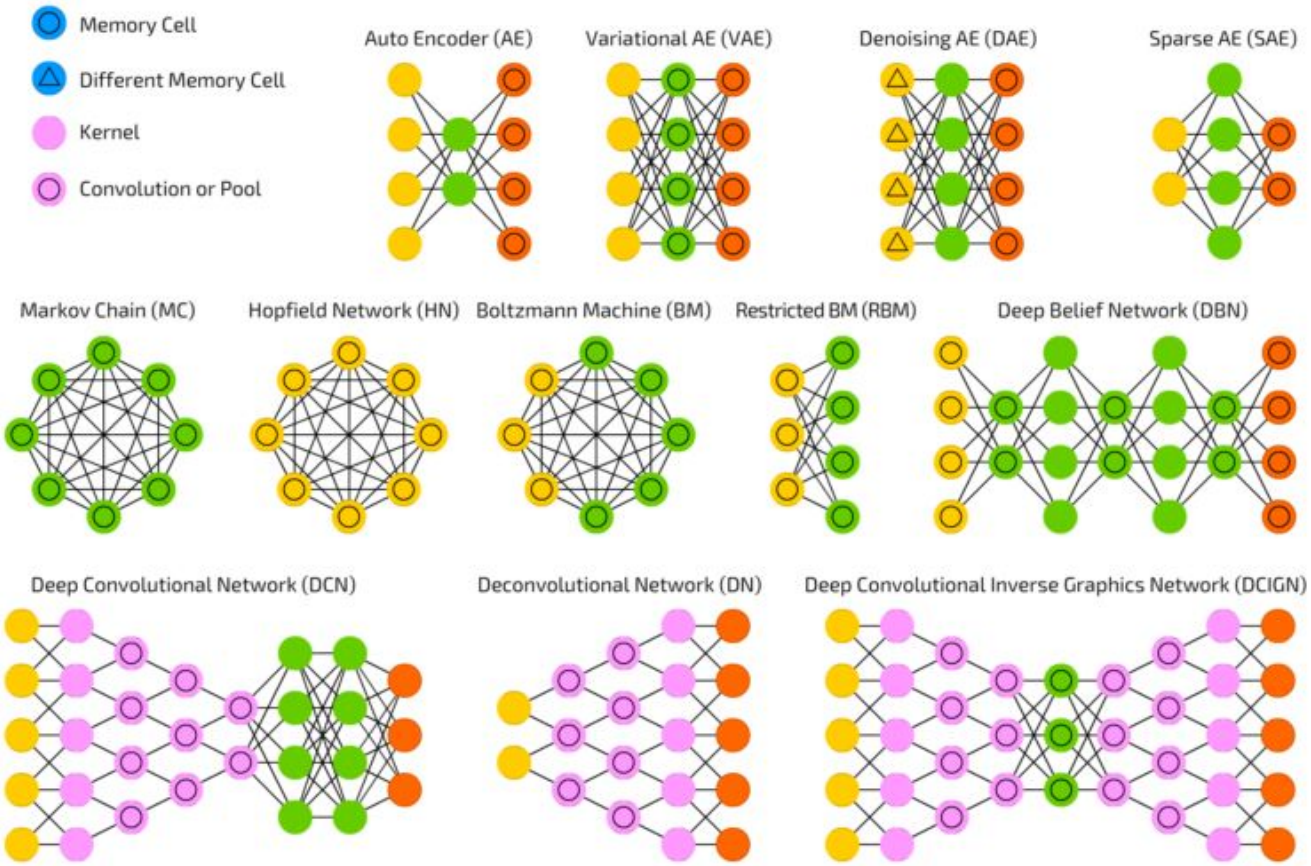
Redes Neurais Artificiais - Backpropagation



Redes Neurais Artificiais - Epoch (cycle)



Redes Neurais Artificiais - Arquiteturas



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Deep vs Machine Learning

Deep vs Machine Learning

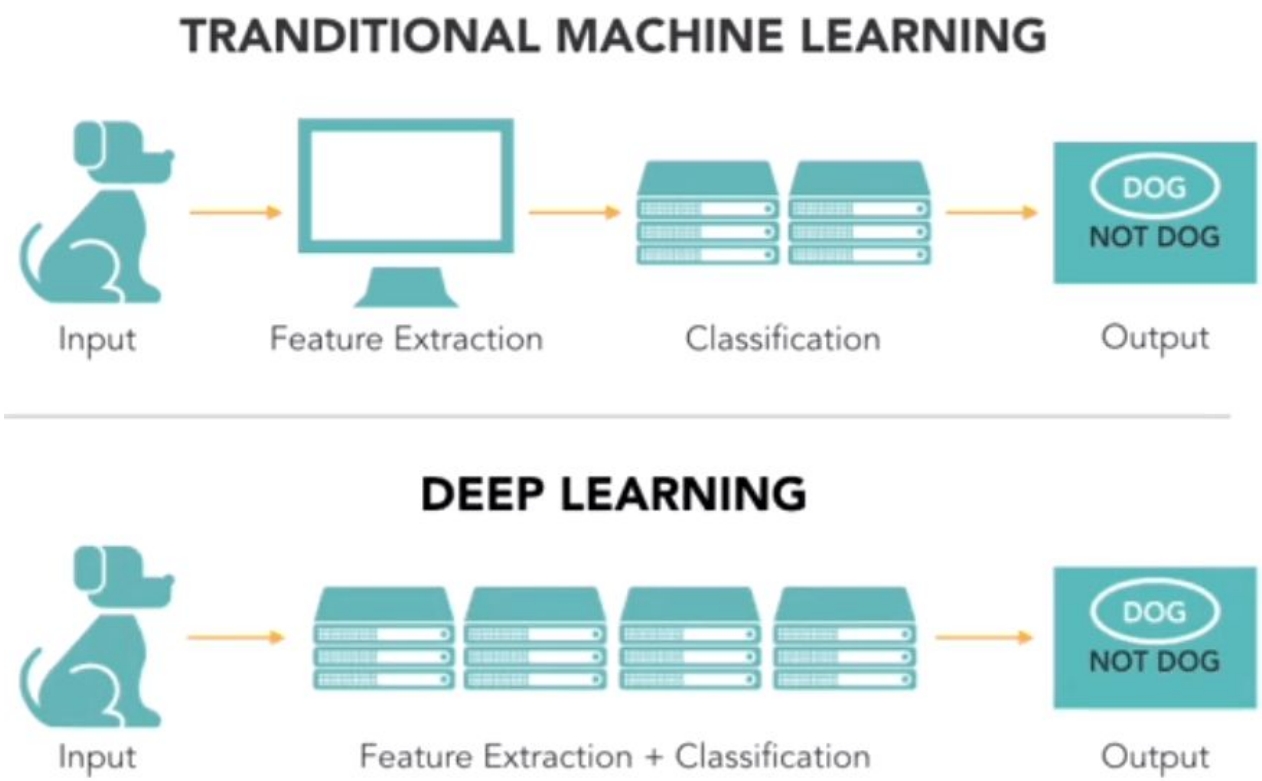


Image Matrix

Atividades de deep learning geralmente envolvem imagens como dados de entrada. Com isso é preciso realizarmos um mapeamento dessa entrada em formato matricial

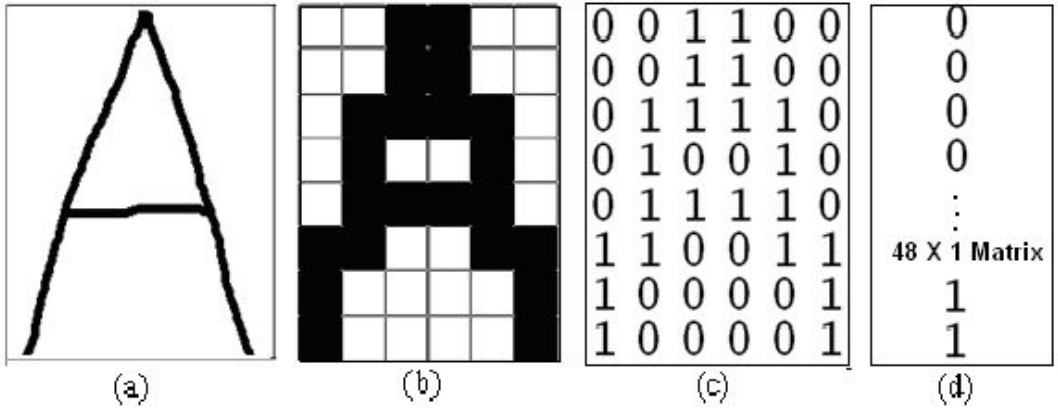
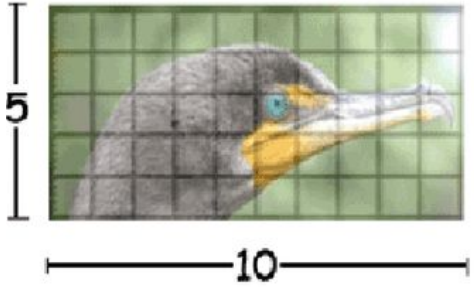
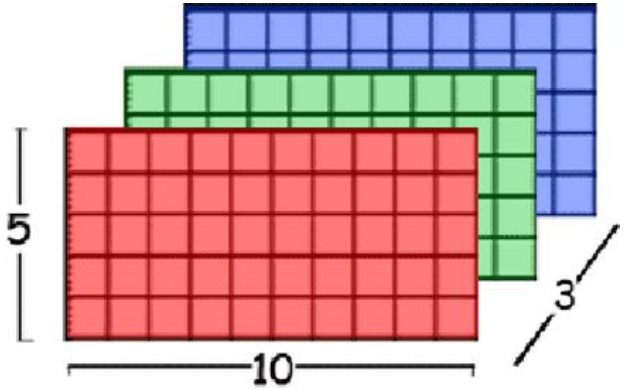


Image Matrix

Já quando tratamos de imagens coloridas precisamos converter em uma única matriz





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Deep Learning

- Ao transformar imagens coloridas em escala de cinza acabamos **por perder algumas informações** que possam ser importantes;
- Para isolar o **objeto de interesse** é preciso realizar uma atividade prévia de **detecção e crop**

Desafio

Em uma aplicação real de identificação de pássaros o contexto da imagem torna a classificação ainda mais difícil

- imagens com alta resolução (**alta dimensionalidade**)
- images com **muitas informações**

Convolução + ReLU

Input		
0	1	2
3	4	5
6	7	8

*

Kernel	
0	1
2	3

=

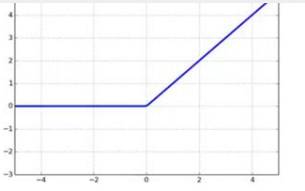
Output	
19	25
37	43

19 = (0*0) + (1*1) + (3*2) + (4*3)

ReLU

0 for x < 0

x for x => 0



9	3	5	-8
-6	2	-3	1
1	3	4	1
3	-4	5	1

→

9	3	5	0
0	2	0	1
1	3	4	1
3	0	5	1

Convolução

O processo de convolução visa maximizar algumas características da imagem de modo que 'novas features' sejam geradas

Input image

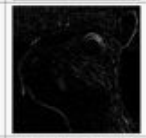
Convolution Kernel

$$\begin{bmatrix} -1 & -1 & -1 \\ -1 & 8 & -1 \\ -1 & -1 & -1 \end{bmatrix}$$

Feature map

Convolução

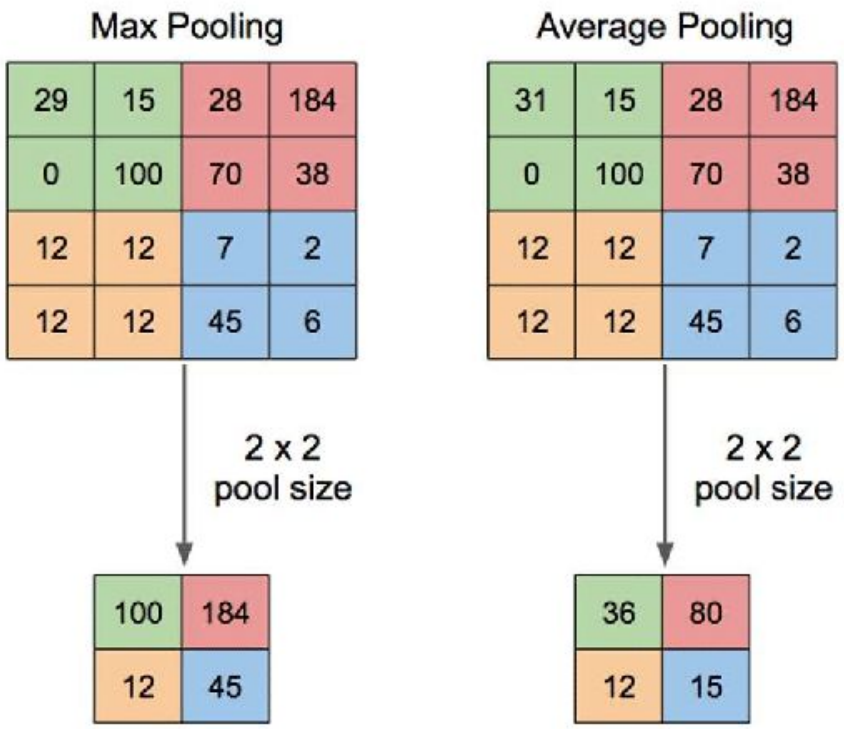
Existem diversas opções de kernel disponível. Cada um com uma finalidade

Operation	Filter	Convolved Image
Identity	$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$	
Edge detection	$\begin{bmatrix} 1 & 0 & -1 \\ 0 & 0 & 0 \\ -1 & 0 & 1 \end{bmatrix}$	
	$\begin{bmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{bmatrix}$	
	$\begin{bmatrix} -1 & -1 & -1 \\ -1 & 8 & -1 \\ -1 & -1 & -1 \end{bmatrix}$	
Sharpen	$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 5 & -1 \\ 0 & -1 & 0 \end{bmatrix}$	
Box blur (normalized)	$\frac{1}{9}\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$	
Gaussian blur (approximation)	$\frac{1}{16}\begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$	

Exemplo prático:
https://www.cs.ryerson.ca/~aharley/vis/conv/

Pooling

Mesmo após a convolução a matriz inicial ainda possui uma **alta dimensionalidade**. Uma abordagem para reduzir o espaço de de valores é através da estratégia do pooling.



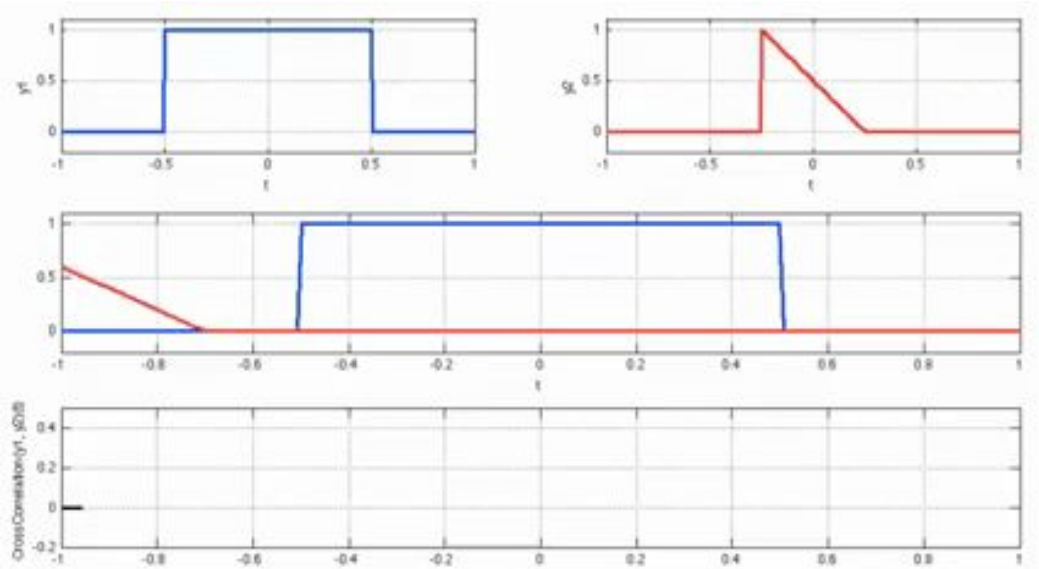
Convolução em outros domínios

Convolução temporal

$$f(t) * g(t) = \int_{-\infty}^{\infty} f(\tau) \cdot g(t-\tau) \cdot d\tau$$

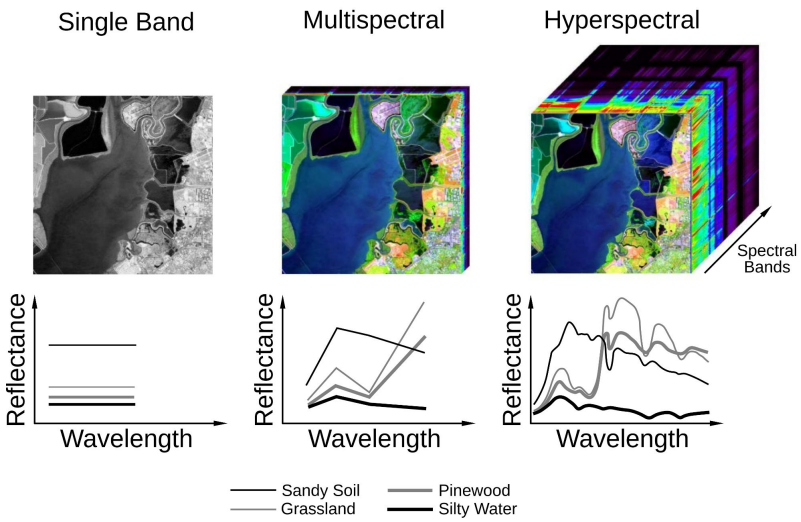
Expressão de convolução temporal contínua

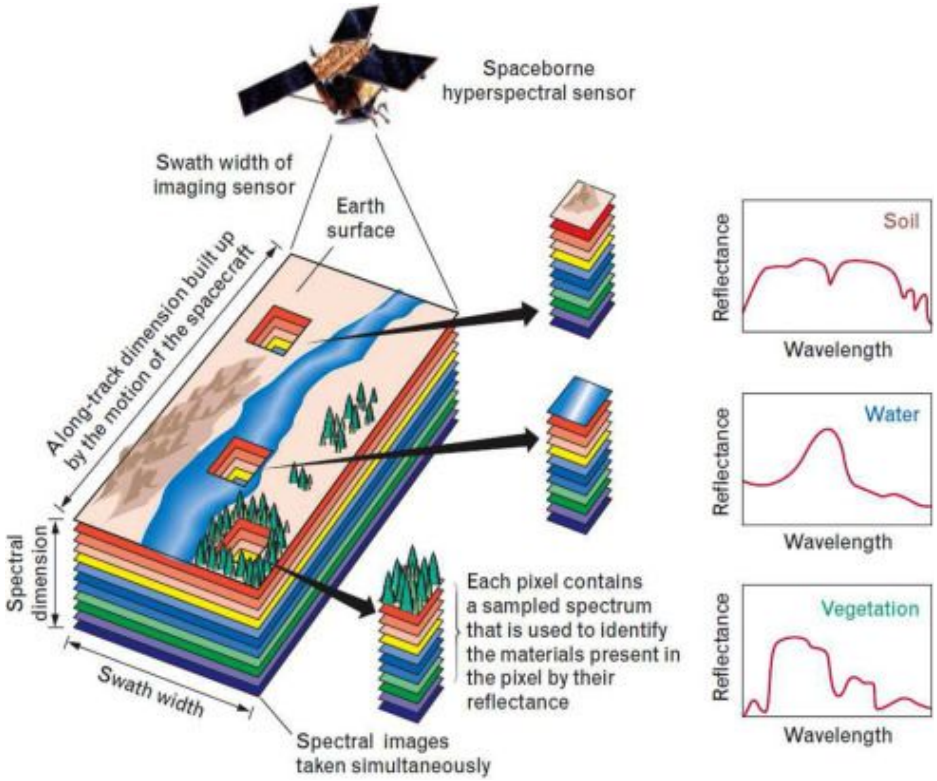
$$(f * g)(n) = \sum_{k=-\infty}^{\infty} f(k) \cdot g(n-k)$$



Convolução em outros domínios

Convolução espectral





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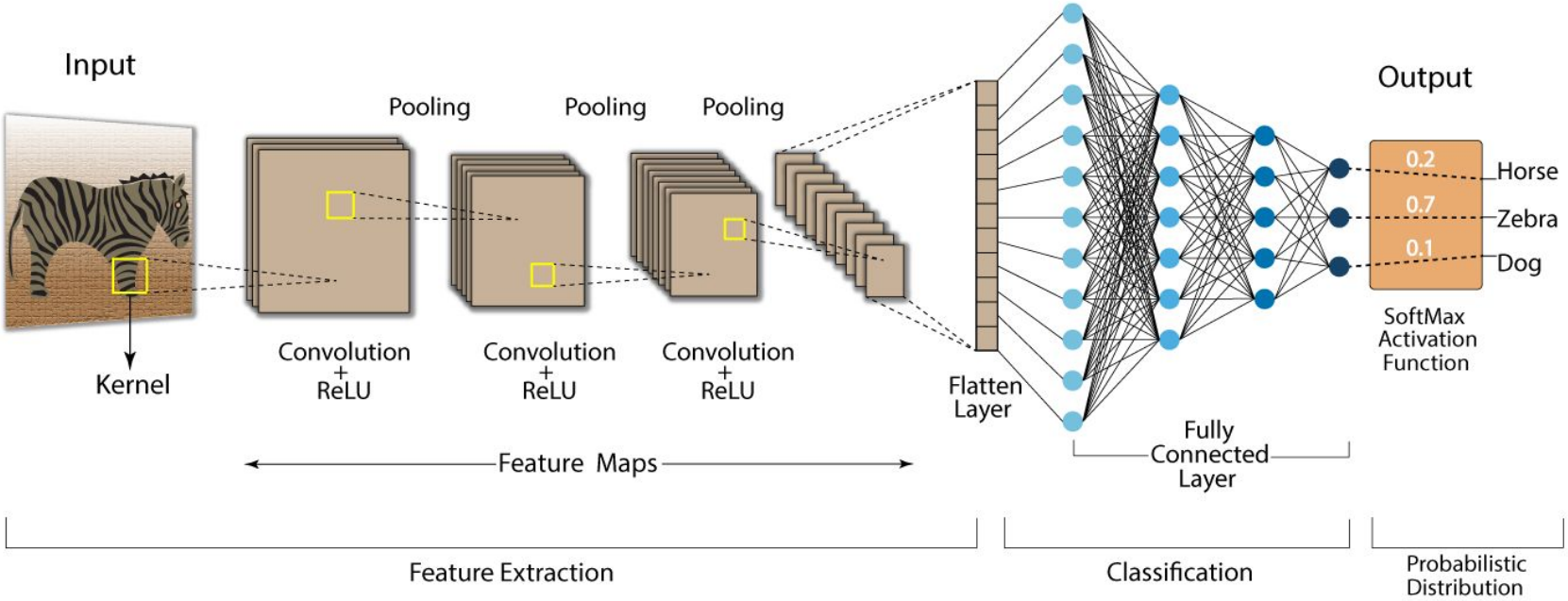
Deep Learning - CNN

- Agora que sabemos como extrair features de imagens podemos aplicar o deep learning a partir de imagens sem nenhuma alteração

Deep Learning - CNN

ref: https://www.kdnuggets.com/2016/11/intuitive-explanation-convolutional-neural-networks.html/3



Exemplo: https://www.tensorflow.org/tutorials/images/cnn

5

Deep Learning - Libraries

Deep Learning - Libraries

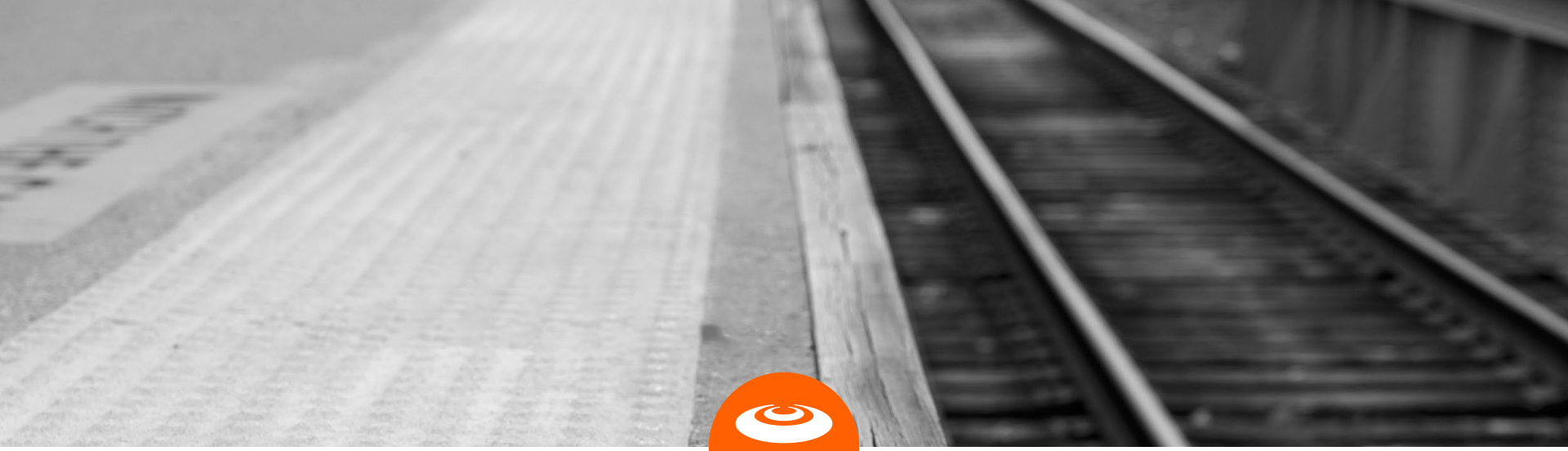
TensorFlow

Keras

PyTorch

scikit learn

python



Hands On!

Experimentos em Machine Learning com Python

Ref: https://www.tensorflow.org/tutorials/images/cnn

C . e . S . A . R

Pessoas impulsionando inovação.

Inovação impulsionando negócios.

NOSSO CONTATO

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